LEC28 - Comparison Tests

Tuesday, March 18, 2025 Section 5.4

Warm Up Problem

Suppose we know that $0 \le a_n \le \frac{1}{3^n}$ for all $n \ge 1$. What can we say about $\sum_{n=1}^{\infty} a_n$?

$\Rightarrow 0 \le \sum_{n=1}^{\infty} a_n \le \sum_{n=1}^{\infty} \frac{1}{3^n}$

$\Rightarrow$ Since $\sum_{n=1}^{\infty} \frac{1}{3^n}$ converges, by the comparison test we have that $\sum_{n=1}^{\infty} a_n$ must also converge

Comparison Test

Let $\{a_n\}$ and $\{b_n\}$ be sequences. Suppose there exists an integer N such that $0 \le a_n \le b_n$ for all $n \ge N$

i) If the larger one $\sum_{n=1}^{\infty} b_n$ converges, then the smaller one $\sum_{n=1}^{\infty} a_n$ converges

ii) If the smaller one $\sum_{n=1}^{\infty} a_n$ diverges, then the larger one $\sum_{n=1}^{\infty} b_n$ diverges

Note that this test can only be used with non-negative terms

Example

Does $\sum_{n=1}^{\infty} \frac{1}{2^n + \sqrt{n}}$ converge or diverge?

$\Rightarrow$ We know that $0 \le \frac{1}{2^n + \sqrt{n}} \le \frac{1}{2^n}$

$\Rightarrow$ Therefore, by the comparison test, since $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges, we conclude that $\sum_{n=1}^{\infty} \frac{1}{2^n + \sqrt{n}}$ must also converge

Example

Does $\sum_{n=1}^{\infty} \frac{4 + \cos(n)}{n^3 + 1}$ converge or diverge?

$\Rightarrow$ We know that $0 \le \frac{4 + \cos(n)}{n^3 + 1} \le \frac{4 + \cos(0)}{n^3} = \frac{5}{n^3}$

$\Rightarrow$ Therefore, by the comparison test, since $\sum_{n=1}^{\infty} \frac{5}{n^3}$ converges, we have that $\sum_{n=1}^{\infty} \frac{4 + \cos(n)}{n^3 + 1}$ must also converge

Limit Comparison Test (LCT)

Let $\{a_n\}$ and $\{b_n\}$ be sequences with non-negative terms

i) If $\lim_{n \to \infty} \left[\frac{a_n}{b_n}\right] = L \ne 0$, then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge

ii) If $\lim_{n \to \infty} \left[\frac{a_n}{b_n}\right] = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges

iii) If $\lim_{n \to \infty} \left[\frac{a_n}{b_n}\right] = \infty$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges

Example

Does $\sum_{n=1}^{\infty} \frac{n-1}{4n^2}$ converge or diverge?

$\Rightarrow$ Choose $a_n = \frac{n-1}{4n^2}$, $b_n = \frac{n}{n^2} = \frac{1}{n}$

$\Rightarrow \lim_{n \to \infty} \left[\frac{a_n}{b_n}\right] = \lim_{n \to \infty} \left[\frac{n-1}{4n^2} \cdot n\right] = \lim_{n \to \infty} \left[\frac{n^2 - n}{4n^2}\right] = \frac{1}{4}$

$\Rightarrow$ Therefore, since $\lim_{n \to \infty} \left[\frac{a_n}{b_n}\right] = L \ne 0$, then $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ both converge or both diverge

$\Rightarrow$ Therefore, since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, $\sum_{n=1}^{\infty} \frac{n-1}{4n^2}$ must also diverge